ECE4063 IC Design Project

Contribution Statement - AES-128 Hardware Accelerator

Item	Details
Project title	AES-128 Hardware Accelerator: Quartus, ModelSim, and Tiny Tapeout Workflow
Group ID	MSMC
Group members	Chin Wei Chun and Foong Yao Le
Contribution distribution	50% Chin Wei Chun / 50% Foong Yao Le

This contribution statement records the main tasks completed by each group member. Although the work was divided by technical area, both members contributed substantially to the final AES-128 project package. The final contribution is declared as 50/50 because the workload was balanced across RTL development, simulation, Quartus analysis, Tiny Tapeout backend work, report writing, verification evidence, and final formatting.

Table B.1: Contribution Summary

Member	Main responsibilities	Estimated contribution
Chin Wei Chun	• Completed the full Tiny Tapeout preparation and backend workflow for both the baseline and optimized AES-128 submissions. • Designed and implemented the optimized RTL architecture, including the low-area shared-resource AES datapath. • Obtained credible AES-128 encryption test vectors used to verify the AES designs. • Prepared major report sections including the executive summary, conclusion, report template, Section 4.0, Section 5.0 to 5.3, Section 6.0, Section 8.0, Section 9.0, citations, and contribution/academic integrity content. • Supported integration, result interpretation, and final project documentation.	50%
Foong Yao Le	• Designed and implemented the baseline RTL architecture and AES transformation modules. • Performed module-level simulations for the AES building blocks and collected simulation evidence. • Completed Quartus compilation and obtained analysis results, including synthesis, resource utilisation, and timing evidence. • Prepared the remaining report sections not assigned above, including background, architecture, implementation details, Quartus evidence, engineering discussion, limitations, submission organisation, and appendices. • Reformatted the report, improved presentation quality, and made the final document more consistent and professional.	50%

Table B.2: Detailed Task Allocation

Task area	Primary contributor	Specific contribution
Optimized RTL design	Chin Wei Chun	Implemented the low-area optimized AES-128 architecture using a shared-resource multi-cycle datapath. This included the optimized control sequencing, shared S-box usage, one-column MixColumns reuse, internal counters, temporary storage, and ciphertext output handling.
Baseline RTL design	Foong Yao Le	Implemented the baseline AES-128 RTL architecture using a clearer one-round-per-cycle iterative structure. This included the baseline wrapper, AES core, round datapath, SubBytes, ShiftRows, MixColumns, AddRoundKey, S-box, and key expansion modules.
AES test vector collection	Chin Wei Chun	Obtained credible AES-128 encryption test vectors from recognised sources and prepared them for use in functional verification of the AES-128 encryption cores.
Module-level simulation	Foong Yao Le	Simulated the AES functional modules individually, including the main AES transformation blocks, to confirm that the building blocks behaved correctly before full-core integration.
Full-core verification support	Both members	Both members contributed to checking that the baseline and optimized AES-128 designs produced the correct ciphertext outputs using known test vectors and self-checking simulation evidence.
Quartus compilation and analysis	Foong Yao Le	Compiled the AES-128 designs in Quartus, obtained synthesis/resource/timing information, and collected the Quartus evidence used in the report.

Task area	Primary contributor	Specific contribution
Tiny Tapeout backend workflow	Chin Wei Chun	Prepared and completed the Tiny Tapeout workflow for both the baseline and optimized designs. This included repository setup, wrapper adaptation, byte-loading interface handling, tile allocation adjustment, hardening workflow execution, GDS/layout evidence, and backend result collection.
Report template and structure	Chin Wei Chun	Prepared the initial report structure, section layout, table format, and overall reporting template used to organise the AES-128 design, verification, Quartus, backend, and trade-off evidence.
Report writing - assigned sections	Chin Wei Chun	Wrote or substantially prepared the executive summary, conclusion, Section 4.0 Engineering Design Choices, the first half of Section 5 before Section 5.4, Section 6.0 Functional Verification, Section 8.0 Tiny Tapeout Backend Workflow Evidence, Section 9.0 Performance Comparison and Optimization Analysis, citation content, and this contribution statement.
Report writing - remaining sections	Foong Yao Le	Wrote or substantially prepared the remaining report sections not listed under Chin Wei Chun, including the introductory/background material, architecture explanation, later RTL implementation sections, Quartus evidence, engineering discussion, limitations, future improvements, submission package organisation, and supporting appendix content.
Report formatting and presentation	Foong Yao Le	Improved document appearance, consistency, formatting, table presentation, layout, section readability, and general report polish so that the final submission was easier to follow.
Final checking and submission readiness	Both members	Both members reviewed the final report, evidence, results, and project package to ensure that the submitted material reflected the completed AES-128 implementation and verification work.

Table B.3: Shared Responsibilities

Shared area	Description
Debugging and integration	Both members helped identify and resolve implementation, simulation, and documentation issues during project development.
Verification interpretation	Both members reviewed simulation outputs and AES ciphertext matching results to confirm that the RTL implementations behaved correctly.
Result discussion	Both members contributed to interpreting the design trade-offs between area, timing, latency, modularity, verification effort, and backend suitability.
Demonstration preparation	Both members prepared to explain the AES-128 architecture, testbench behaviour, Quartus results, and Tiny Tapeout backend outputs during assessment.
Final project package	Both members contributed to ensuring that RTL files, testbenches, simulation evidence, Quartus reports, Tiny Tapeout evidence, report material, citations, and supporting documents were ready for submission.

Declaration of Equal Contribution

The group declares that the final contribution is 50% Chin Wei Chun and 50% Foong Yao Le. The distribution is considered equal because Chin Wei Chun primarily handled the optimized RTL, Tiny Tapeout backend workflow, test vector sourcing, and several major report sections, while Foong Yao Le primarily handled the baseline RTL, module-level simulation, Quartus compilation and analysis, remaining report sections, and final report formatting. Both members also contributed to debugging, verification interpretation, engineering discussion, and final submission preparation.

Sign-off

Member	Signature	Date
Chin Wei Chun	Wei Chun	29/5/2026
Foong Yao Le	Yao Le	29/5/2026